

Micro Energy Harvesting 2026

Exercise 1: Energy, power, energy forms and conversion

We want to start slowly, so this first exercise will introduce you to forms of energy, expected amounts of energy, power and power consumption.

Task 1: Energy

What was energy again?

Oh yes, there is something like potential energy, kinetic energy (& rotational energy), chemical energy, thermal energy, electrical energy, relativistic energy, quantum mechanical energy, etc.

- a) Identify the formulas for the above-mentioned forms of energy (if possible) and write them down together with an example and a typical energy value.
- b) How are energy and power related?
- c) If you remember back to the first semester Physics. How much energy is expended when you lift a mass of 1 kg by one meter?
- d) And - let's assume you lose this energy again when you lower the mass again - how much power do you generate when you lift it every 10 seconds?
- e) Calculate the physical performance of a person with a body weight of 75 kg who jumps 10 cm into the air twice per second during boxing training (rope skipping). Also calculate the total energy consumption if you keep this up for 30 minutes.

Task 2: Energy content of materials, thermal energy, grey energy

In the first lecture the manufacturing energy for a copper cable with PVC sheathing was calculated and compared to the energy demand of a typical sensor. Comparing is fun, so you can compare some other things energetically!

- a) Look up the energy content of the following things from literature: (The unit of energy is Joule J)

Heating value:

- Small serving of chips
- 1 L Diesel
- 1 L Ethanol
- 1 kg Hard coal
- 1 stere of air-dry (15 % residual moisture) beech firewood
- Gas
- Hydrogen - H₂

- b) Find the energy required to heat the following items from 20 °C (293 K) to the specified temperature.

Thermal energy:

- 1 drop (=0.05 ml) of water, heated to 30 °C
- 1 drop of alcohol, heated to 30 °C
- 1 drop of copper at melting point
- 1 drop of aluminum at melting point
- 1 drop of tungsten at melting point

- c) Determine the energy required to produce the following substances (difficult for some).

Gray (Embodied) energy:

- 1 kg silicon, wafer grade
- 1 kg aluminium
- 1 kg copper
- 1 kg gold
- 1 kg glass (e.g. drinking glass)
- 1 L beer (Pils)
- 1 car
- 1 pound of coffee
- 1 € (per issued)

Task 3: Typical input power for some devices

Estimate the power consumption of the following technical devices by internet research (additional task: try to estimate the values in advance - note the difference!)

Typical input power:

- LED green, super bright @operating voltage
- CREE LED, MK-R, neutral white
- Bike light (no matter where mounted / illuminant)
- Vacuum cleaner
- Smartphone (during transmission, sending and display)
- Cortex M0+ microprocessor at 20 MHz clock frequency
- MSP 430 μ C with FRAM (5529) in deep sleep mode
- Arduino Uno at 5 V under full load